Solution Tutorial II – Drag

Task 1

Calculate the airspeed V at $M = 0.8$ and determine the dynamic viscosity μ as well as the density ρ of air at an altitude of $h = 10km$.

- Airspeed:

$$V = M \cdot a = 0.8 \cdot 299.4 = 239.5\ m/s$$

- Dynamic viscosity (either calculation or value from ISA table):

$$\mu = 18.27 \cdot 10^{-6} \cdot \left(\frac{411.15}{T_{(°K)}+120}\right) \cdot \left(\frac{T_{(°K)}}{291.15}\right)^{1.5} \text{ (Equation of Sutherland)}$$

$$\Longrightarrow \mu = 18.27 \cdot 10^{-6} \cdot \left(\frac{411.15}{223.15 + 120}\right) \cdot \left(\frac{223.15}{291.15}\right)^{1.5} = 14.69 \cdot 10^{-6} N \cdot s/m^2$$

- Air density (value from ISA table):

$$\rho = 0.412\ kg/m^3$$

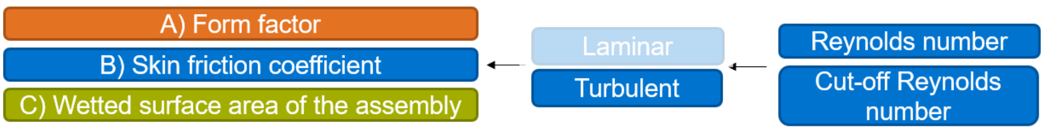
Task 2

Determine the equivalent parasite drag areas f_j of the individual assemblies (fuselage, wings, empennage and engines) at the flight conditions of task 1). The assemblies are constructed of polished sheet. Neglect interference drag and assume fully turbulent flow over all assemblies.

- General approach:

$$D_0 = \frac{1}{2}\rho V^2 \cdot S_{ref} \cdot C_{D0} = \frac{1}{2}\rho V^2 \cdot \sum_{j=1}^{n} \underbrace{S_j \cdot C_{D0,j}}_{\text{Equivalent } component \text{ parasite drag } \mathbf{area}}$$

$$f_j = \boxed{F_j} \cdot \boxed{c_{f,j}} \cdot \boxed{S_j} = C_{D0,j} \cdot S_{Ref}$$



- Fuselage:

A) Form factor

$$F_{fus} = \left(0.9 + \frac{5}{f^{1.5}} + \frac{f}{400}\right) \text{ with } f = \left(\frac{l}{d}\right)_{fus}$$

Fuselage diameter:

$$d = \frac{1}{2} \cdot (w_{fus} + H_{fus}) = \frac{1}{2} \cdot (3.95 + 4.14) = 4.045m$$

$$\Longrightarrow F_{fus} = \left(0.9 + \frac{5}{\left(\frac{37.57}{4.045}\right)^{1.5}} + \frac{\frac{37.57}{4.045}}{400} \right) = \mathbf{1.100}$$

B) Skin friction coefficient

Reference length:

$$l_{fus} = 37.57\ m$$

Reynolds number:

$$Re_{fus} = \frac{\rho \cdot V \cdot l_{fus}}{\mu} = \frac{0.412 \cdot 239.5 \cdot 37.57}{14.69 \cdot 10^{-6}} = \mathbf{2.524 \cdot 10^{8}} \rightarrow \text{turbulent air flow}$$

Cut-off Reynolds number (transsonic equation, as $M = 0.8$):

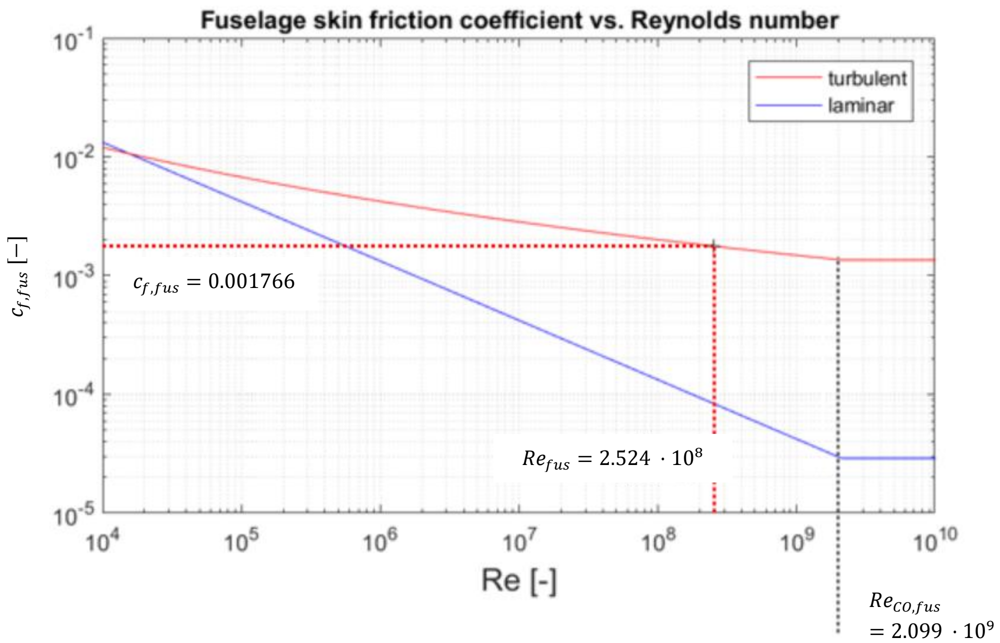
$$Re_{CO,\mathrm{fus}} = 44 \cdot \left(\frac{l_{fus}}{k}\right)^{1.053} \cdot M_{\infty}^{1.16} = 44 \cdot \left(\frac{37.57}{1.5 \cdot 10^{-6}}\right)^{1.053} \cdot 0.8^{1.16} = \mathbf{2.099 \cdot 10^{9}}$$

Proceed with the smallest of Re_{fus} and $Re_{CO,fus}$. In this case: Re_{fus}

Skin friction coefficient:

$$c_{f,fus} = \frac{0.455}{(\log_{10} Re_{fus})^{2.58}} \cdot \left(1 + 0.144 \cdot {M_{\infty}}^{2}\right)^{-0.65}$$

$$\Longrightarrow c_{f,fus} = \frac{0.455}{(\log_{10} 2.524 \cdot 10^{8})^{2.58}} \cdot (1 + 0.144 \cdot 0.8^{2})^{-0.65} = \mathbf{0.001766}$$



C) Wetted surface area of the assembly

Fuselage diameter as above (mean of fuselage width and height)

Plate analogy area:

$$S_{fus,wet} = \pi \cdot d^2 \cdot \left[\left(\frac{l}{d}\right)_{fus} - 1.35\right]$$

$$\Longrightarrow S_{fus,wet} = \pi \cdot 4.045^2 \cdot \left[\left(\frac{37.57}{4.045}\right)_{fus} - 1.35\right] = \mathbf{408\ m^2}$$

Result of Fuselage parasite drag area:

$$\boldsymbol{f_{fus}} = c_{f,fus} \cdot F_{fus} \cdot S_{fus,wet} = 0.001766 \cdot 1.100 \cdot 408.0 = \mathbf{0.7926\ m^2}$$

- Wing:

A) Form factor

Location of maximum airfoil thickness: $(x/c)_m = 39.90\%$; $\Lambda_m = 25.78°$

Form factor:

$$F_W = \left[1 + \frac{0.6}{(x/c)_m} \cdot \left(\frac{t}{c}\right)_W + 100 \cdot \left(\frac{t}{c}\right)_W^4\right] \cdot \left[1.34\ M^{0.18}\ (\cos \Lambda_m)^{0.28}\right]$$

$$\Longrightarrow F_W = \left[1 + \frac{0.6}{39.90\%} \cdot (11.75\%)_W + 100 \cdot (11.75\%)_W^4\right]$$
$$\cdot \left[1.34\ 0.8^{0.18}\ (\cos 25.78)^{0.28}\right] = \mathbf{1.495}$$

B) Skin friction coefficient

Reference length:

$$c_{MAC} = 4.1935\ m$$

Reynolds number:

$$Re_W = \frac{\rho \cdot V \cdot c_{MAC}}{\mu} = \frac{0.412 \cdot 239.5 \cdot 4.1935}{14.69\ 10^{-6}} = \mathbf{2.817 \cdot 10^7} \rightarrow \text{turbulent air flow}$$

Cut-off Reynolds number (transsonic equation, as $M = 0.8$):

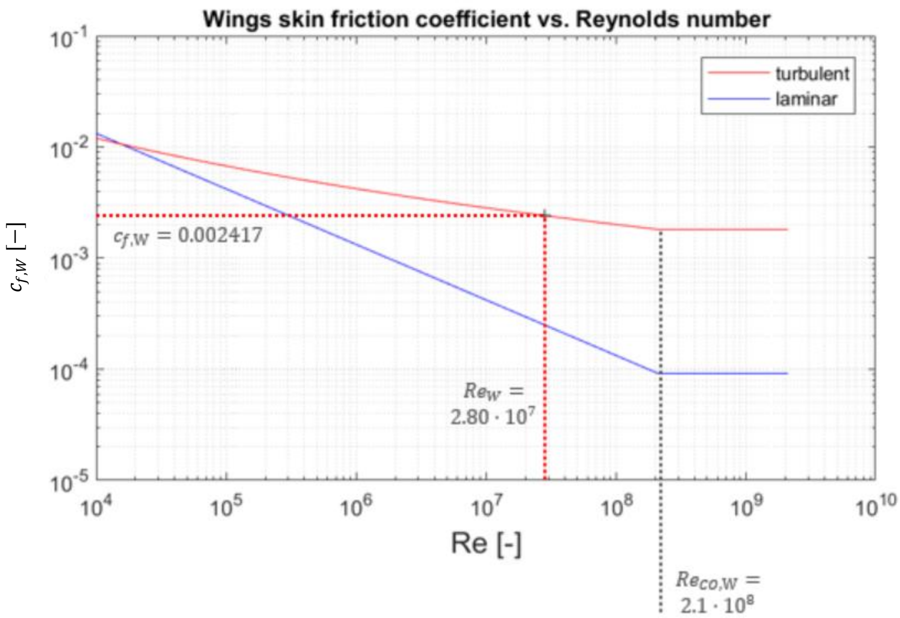
$$Re_{CO,\mathrm{W}} = 44 \cdot \left(\frac{c_{MAC}}{k}\right)^{1.053} \cdot M_\infty^{1.16} = 44 \cdot \left(\frac{4.1935}{1.5 \cdot 10^{-6}}\right)^{1.053} \cdot 0.8^{1.16} = \mathbf{2.085 \cdot 10^8}$$

Proceed with the smallest of Re_W and $Re_{CO,\mathrm{W}}$. In this case: Re_W

Skin friction coefficient:

$$c_{f,\mathrm{W}} = \frac{0.455}{(\log_{10} Re_\mathrm{W})^{2.58}} \cdot \left(1 + 0.144 \cdot {M_\infty}^2\right)^{-0.65}$$

$$\Longrightarrow c_{f,\mathrm{W}} = \frac{0.455}{(\log_{10} 2.817 \cdot 10^7)^{2.58}} \cdot (1 + 0.144 \cdot 0.8^2)^{-0.65} = \mathbf{0.002415}$$



C) Wetted surface area of the assembly

Plate analogy area:

$$S_{W,wet} = 2 \cdot S_{ref} = 2 \cdot 122.4 = \mathbf{244.8\ m^2}$$

Result of Wing parasite drag area:

$$\mathbf{f_W} = \mathrm{c_{f,W}} \cdot \mathrm{F_W} \cdot \mathrm{S_{W,wet}} = 0.002415 \cdot 1.495 \cdot 244.8 = \mathbf{0.8838\ m^2}$$

- Empennage (simple approach as in lecture notes):

$$\mathrm{f_{EMP}} = \mathrm{C_{D,EMP}} \cdot (S_{HT} + S_{VT}) = 0.007 \cdot (31 + 21.5) = \mathbf{0.3675\ m^2}$$

- Nacelle:

A) Form factor

$$F_N = \left[1 + {0.35}/{f}\right] \text{ with } f = \left(\frac{l}{d}\right)_{nac}$$

$$\Rightarrow F_{nac} = \left[1 + {0.35}\Big/{\frac{4.44}{2.37}}\right] = \mathbf{1.187} \text{ (assume engine size = nacelle size)}$$

B) Skin friction coefficient

Reference length:

$$l_{nac} = 4.44\ m$$

Reynolds number:

$$Re_{nac} = \frac{\rho \cdot V \cdot l_{nac}}{\mu} = \frac{0.412 \cdot 239.5 \cdot 4.44}{14.69 \cdot 10^{-6}} = \mathbf{2.982 \cdot 10^7} \rightarrow \text{turbulent air flow}$$

Cut-off Reynolds number (transsonic equation, as $M = 0.8$):

$$Re_{CO,nac} = 44 \cdot \left(\frac{l_{nac}}{k}\right)^{1.053} \cdot M_\infty^{1.16} = 44 \cdot \left(\frac{4.44}{1.5 \cdot 10^{-6}}\right)^{1.053} \cdot 0.8^{1.16} = \mathbf{2.215 \cdot 10^8}$$

Proceed with the smallest of Re_{nac} and $Re_{CO,nac}$. In this case: Re_{nac}

Skin friction coefficient:

$$c_{f,\text{nac}} = \frac{0.455}{(\log_{10} Re_{nac})^{2.58}} \cdot \left(1 + 0.144 \cdot M_\infty{}^2\right)^{-0.65}$$

$$\Rightarrow c_{f,\text{nac}} = \frac{0.455}{(\log_{10} 2.982 \cdot 10^7)^{2.58}} \cdot (1 + 0.144 \cdot 0.8^2)^{-0.65} = \mathbf{0.002395}$$

C) Wetted surface area of the assembly

$$S_{nac,wet} = \pi\, d_{nac} l_{nac} = \pi \cdot 2.37 \cdot 4.44 = \mathbf{33.06}\, \boldsymbol{m}^2$$

Result of Nacelle parasite drag area:

Do not forget to multiply the formula in the lecture notes with the number of engines:

$$\boldsymbol{f_{nac}} = c_{f,nac} \cdot F_{nac} \cdot S_{nac,wet} \cdot n_{eng} = 0.002395 \cdot 1.187 \cdot 33.06 \cdot \mathbf{2} = \mathbf{0.1880\ m^2}$$

Task 3

Now derive the total parasite drag area f from the individual parasite drag areas previously calculated. Finally, derive the zero-lift drag coefficient C_{D0} of the Airbus A320 from this value.

- Total parasite drag area:

$$f = f_{fus} + \mathrm{f_W} + \mathrm{f_{EMP}} + f_{nac} = 0.7926 + 0.8838 + 0.3675 + 0.1880 = \mathbf{2.232}\, \boldsymbol{m}^{\mathbf{2}}$$

- Aircraft zero-lift drag coefficient:

$$D_0 = \frac{1}{2}\rho V^2 \cdot C_{D0} \cdot S_{ref} = \frac{1}{2}\rho V^2 \cdot \sum f_n = \frac{1}{2}\rho V^2 \cdot f$$

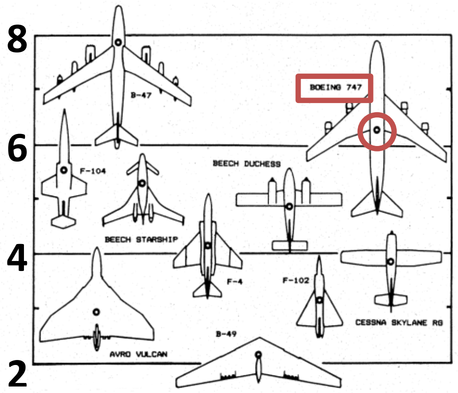
$$\Rightarrow \mathrm{C_{D0}} = {}^{f}\!/_{S_{ref}} = \frac{2.232}{122.4} = \mathbf{0.01824}$$

Similarly, with the equivalent skin friction method (was not asked for in this task):

$$\mathrm{C_{D0}} = c_{fe} \frac{S_{wet}}{S_{ref}} = 0.0026 \cdot 6.3 = 0.01638 \text{ (very rough estimation!)}$$

Aircraft class	c_{fe}
civil transport	0.0026
general aviation, single engine	0.0055
general aviation, twin engine	0.0045
seaplane, piston engine	0.0065
seaplane, jet engine	0.0040

$\frac{S_{wet}}{S_{ref}}$

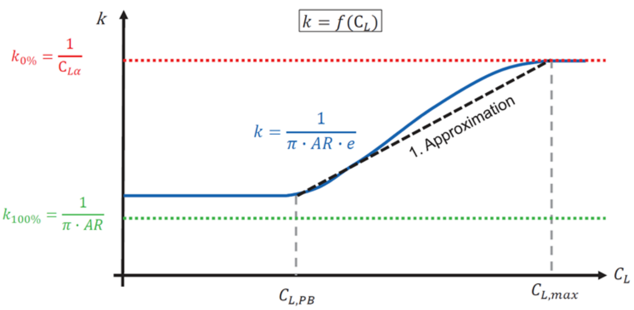


Task 4

Determine the relation between the coefficient of induced drag C_{Di} and the lift coefficient C_L for this aircraft in clean configuration. Also, draw the drag polar, C_D vs. C_L (Note$C_{L,max}$ 1.5 assumed)

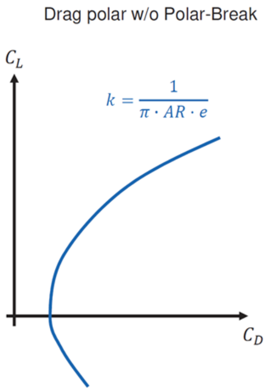
Recap:

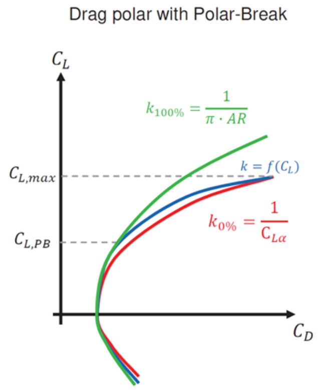
$C_{Di} = k \cdot C_L^2$ ➡ Case differentiation
- Small/moderate C_L ($C_L \leq C_{L,PB}$)
- Combined ($C_{L,PB} \leq C_L < C_{L,max}$) With $C_{L,PB} = 6.5 \cdot \frac{t}{c}$
- Large C_L ($C_L > C_{L,max}$)



for $C_L > C_{L,PB}$ yields:
(flat wings)

$$k = \frac{(C_L - C_{L,PB})}{(C_{L,max} - C_{L,PB})} \cdot \left(\frac{1}{C_{L\alpha}} - \frac{1}{\pi \cdot AR \cdot e}\right) + \frac{1}{\pi \cdot AR \cdot e}$$





Consider the polar break starting at $C_{L,PB}$:

Approximation of $C_{L,PB}$: $C_{L,PB} = 6.5 \cdot \frac{t}{c} = 6.5 \cdot 11.75\% = \mathbf{0.7638}$

Case differentiation:

- For $C_L \leq C_{L,PB}$ (small/moderate C_L):

$$C_{Di} = k \cdot C_L^2, \text{ with } k = \frac{1}{\pi \cdot AR \cdot e}$$

$$C_{Di} = \frac{1}{\pi \cdot 9.39 \cdot 0.85} \cdot C_L^2 = \mathbf{0.03988 \cdot C_L^2}$$

- For $C_L > C_{L,max}$ (large C_L) – for the sake of completeness, although just theoretical:

$$C_{Di} = k \cdot C_L^2, \text{ with } k = \frac{1}{C_{L\alpha}}$$

Calculate $C_{L\alpha}$ with Polhamus ($M = 0.8$, see Tutorial 1):

$$C_{L\alpha} = \frac{2\pi \cdot AR}{2 + \sqrt{\frac{AR^2 \cdot \beta^2}{K^2} \cdot \left(1 + \frac{tan^2 \Lambda_{50\%}}{\beta^2}\right) + 4}}$$

$$= \frac{2\pi \cdot 9.39}{2 + \sqrt{\frac{9.39^2 \cdot 0.6^2}{1.088^2} \cdot \left(1 + \frac{0.4011^2}{0.6^2}\right) + 4}} = 6.907 \text{ /rad}$$

$$\Longrightarrow C_{Di} = \frac{1}{6.91} \cdot C_L^2 = \mathbf{0.1448 \cdot C_L^2}$$

- For $C_{L,PB} \leq C_L < C_{L,max}$ (combined)

$$k = \frac{(C_L - C_{L,PB})}{(C_{L,max} - C_{L,PB})} \cdot \left(\frac{1}{C_{L\alpha}} - \frac{1}{\pi \cdot AR \cdot e}\right) + \frac{1}{\pi \cdot AR \cdot e}$$

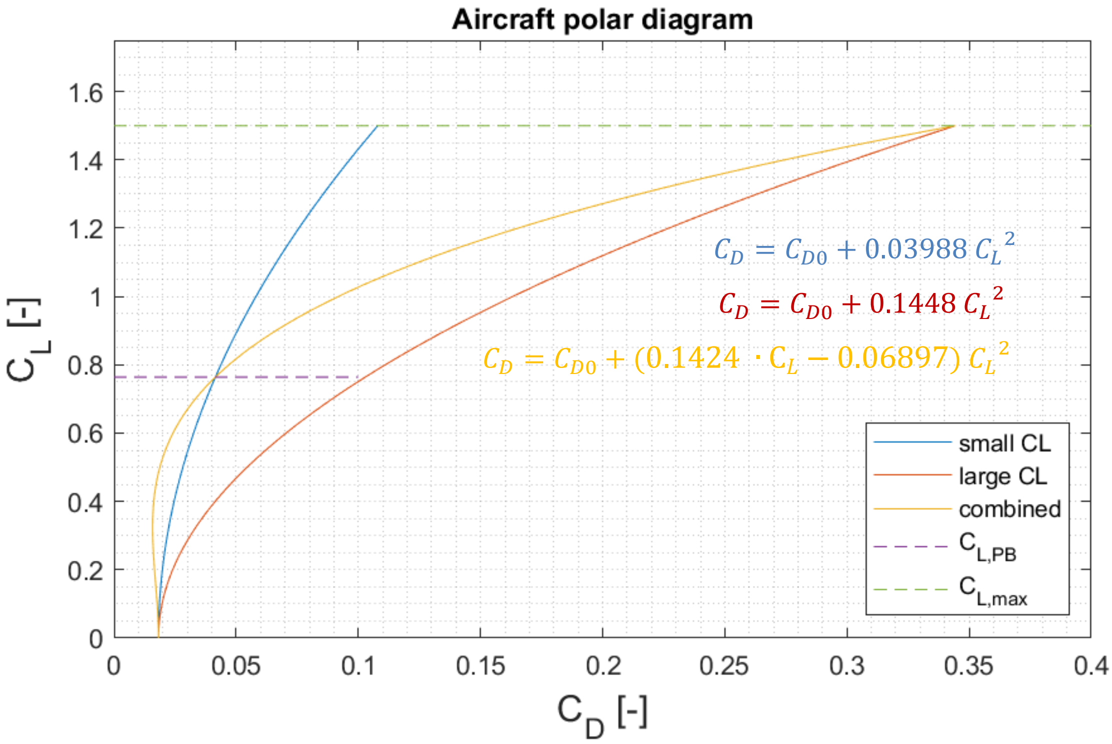
$$= \frac{(C_L - 0.7638)}{(1.5 - 0.7638)} \cdot (0.1448 - 0.03988) + 0.03988$$

$$= 0.1425 \cdot C_L - 0.06897$$

$$\Longrightarrow C_{Di} = (0.1425 \cdot C_L - 0.06897) \cdot C_L^2$$

$$= 0.1425 \cdot C_L^3 - 0.06897\, C_L^2$$

- Drag polar drawing:



Task 5

What thrust force is generated by each engine in cruise (conditions of task 1) ($\sigma = 0°$) at an aircraft mass of $59000\ kg$*?*

Conditions for level flight:

$$L = w \Longrightarrow \frac{1}{2}\rho V^2 \cdot C_L \cdot S_{ref} = w$$

$$\Longrightarrow C_L = \frac{w}{\frac{1}{2}\rho V^2 \cdot S_{ref}} = \frac{59000 \cdot 9.81}{\frac{1}{2} \cdot 0.412 \cdot 239.5^2 \cdot 122.4} = 0.4002$$

$$\Longrightarrow T = \frac{1}{2}\rho V^2 \cdot (C_{D0} + C_{Di}) \cdot S_{ref}$$

$$\Longrightarrow T = \frac{1}{2}\rho V^2 \cdot (C_{D0} + k \cdot C_L^2) \cdot S_{ref}$$

$$\Longrightarrow T = \frac{1}{2}\rho V^2 \cdot \left(C_{D0} + \frac{1}{\pi \cdot AR \cdot e} \cdot C_L^2\right) \cdot S_{ref}$$ (small C_L (< $C_{L,PB}$) in cruise)

$$\Longrightarrow T = \frac{1}{2} \cdot 0.412 \cdot 239.5^2 \cdot (0.01824 + 0.03988 \cdot 0.4002^2) \cdot 122.4 = 35620\ N$$

$$T_{each} = {}^{T}\!/_{2} = {}^{35700}\!/_{2} = \mathbf{17810\ N}$$